1.-18. (canceled)

19. (new) A system for generating recipient-associated distinguishable versions of audio-video content and identifying a recipient associated with a suspected unauthorized distribution of the audio-video content, the system comprising one or more processors and memory storing instructions that, when executed by the one or more processors, cause the system to:

receive video captured from a plurality of cameras and a structured list of edit instructions comprising, for each of a plurality of director-commanded camera cuts, a first edit entry comprising an out-point time code and a first source-camera identifier identifying a first camera of the plurality of cameras as a first source camera, and a following edit entry comprising an in-point time code and a second source-camera identifier identifying a second camera of the plurality of cameras as a second source camera different from the first source camera;

produce reference audio-video content according to the structured list of edit instructions;

generate one or more mates of the reference audio-video content by, for each of a plurality of selected camera cuts among the director-commanded camera cuts, modifying, in the structured list of edit instructions, the out-point time code of the corresponding first edit entry and the in-point time code of the corresponding following edit entry while retaining the first and second source-camera identifiers and the order of the first and second source cameras, such that, for each selected camera cut:

a transition from the first source camera to the second source camera occurs at a reference camera-switch timing in the reference audio-video content and at a later mate camera-switch timing in at least one of the one or more mates; and

in the at least one mate, selection of the first source camera is extended from the reference camera-switch timing to the later mate camera-switch timing and selection of the second source camera begins at the later mate camera-switch timing;

segment an ensemble comprising the reference audio-video content and the one or more mates into chunks;

generate a plurality of manifest files pointing to respective combinations of chunks selected from the ensemble, each respective combination of chunks preserving, for each of the plurality of selected camera cuts, a choice between the reference camera-switch timing and the later mate camera-switch timing of that selected camera cut, so that the respective combinations of chunks represent respective combinations of timing choices across the plurality of selected camera cuts;

cause delivery of streamed audio-video content to respective recipients according to respective manifest files of the plurality of manifest files;

store, in a ledger, associations between the respective manifest files and the recipients to which the streamed audio-video content was delivered according to the respective manifest files;

receive the suspected unauthorized distribution;

apply a scene-change detection algorithm to the suspected unauthorized distribution to identify a plurality of time codes of camera cuts in the suspected unauthorized distribution;

build one or more reconstructed manifest files from the plurality of identified time codes, the one or more reconstructed manifest files including a reconstructed manifest file representing a detected combination of camera-switch timings across the plurality of selected camera cuts; and

search the ledger to identify a recipient associated with a delivered manifest file that is equal to the reconstructed manifest file, the delivered manifest file that is equal to the reconstructed manifest file representing a recipient-associated combination of timing choices that includes at least one of the later mate camera-switch timings and matches the detected combination represented by the reconstructed manifest file.

20. (new) The system of claim 19, wherein the instructions cause the system to preserve, in at least one of the one or more mates, a timing of a later camera cut from the reference audio-video content, thereby restoring synchronization between that mate and the reference audio-video content from the later camera cut onward.

21. (new) The system of claim 20, wherein modifying the out-point and in-point time codes causes the mate camera-switch timing for one of the selected camera cuts to occur ten frames later than the corresponding reference camera-switch timing, extends selection of the first source camera by ten frames, correspondingly shortens a following selection of the second source camera, and restores synchronization at the later camera cut.

22. (new) The system of claim 19, wherein the structured list of edit instructions is an Edit Decision List.

23. (new) The system of claim 19, wherein the video is captured live from diverse viewpoints and the structured list of edit instructions records the director-commanded camera cuts as real-time selections among the plurality of cameras.

24. (new) The system of claim 19, wherein the instructions cause the system to generate a respective mate for each selected camera cut, each respective mate differing from the reference audio-video content in the modified out-point and in-point time codes for only that selected camera cut.

25. (new) The system of claim 19, wherein identifying the plurality of time codes comprises comparing frames of the suspected unauthorized distribution with frames of one or both of the reference audio-video content and the one or more mates using perceptual hashes of the frames.

26. (new) The system of claim 25, wherein the comparing comprises fuzzy matching in which perceptual hashes of groups of frames are compared using sliding windows.

27. (new) The system of claim 19, wherein the chunks are distributed through a content delivery network using adaptive streaming and the plurality of manifest files is tailored to recipient devices or network conditions.

28. (new) The system of claim 19, wherein a mixing process integrates chunks of the reference audio-video content with chunks of the one or more mates and progressively assigns respective manifest files to recipients as additional selected camera cuts become available.

29. (new) The system of claim 19, wherein the suspected unauthorized distribution comprises portions obtained from streamed audio-video content delivered according to different manifest files, and wherein the instructions further cause the system to apply a probabilistic fingerprinting algorithm to recipient-associated sequences of chunk selections represented by the delivered manifest files and identify one or more recipients whose delivered streamed audio-video content contributed respective portions to the suspected unauthorized distribution.

30. (new) The system of claim 29, wherein the probabilistic fingerprinting algorithm comprises a segmented Tardos fingerprinting algorithm that applies respective fingerprints to content segments and identifies at least one contributing recipient for at least one of the respective portions.

31. (new) The system of claim 19, wherein a first manifest file points to a first chunk selected from the reference audio-video content and a second manifest file points to a second chunk selected from one of the one or more mates, the first chunk and the second chunk spanning the same playback interval and having equal playback durations.

32. (new) The system of claim 31, wherein, for one of the selected camera cuts:

the first chunk contains frames from the corresponding first source camera before the reference camera-switch timing and frames from the corresponding second source camera after the reference camera-switch timing; and

the second chunk contains frames from the corresponding first source camera before the mate camera-switch timing and frames from the corresponding second source camera after the mate camera-switch timing.

33. (new) The system of claim 19, wherein causing delivery comprises unicasting respective streamed audio-video content to the respective recipients.

34. (new) The system of claim 19, wherein the instructions further cause one or more audio or video elements not present in the video received from the plurality of cameras to be overlaid onto

at least one of the reference audio-video content or the one or more mates before segmentation into the chunks.

35. (new) A method for generating recipient-associated distinguishable versions of audio-video content and identifying a recipient associated with a suspected unauthorized distribution of the audio-video content, the method comprising:

receiving video captured from a plurality of cameras and a structured list of edit instructions comprising, for each of a plurality of director-commanded camera cuts, a first edit entry comprising an out-point time code and a first source-camera identifier identifying a first camera of the plurality of cameras as a first source camera, and a following edit entry comprising an in-point time code and a second source-camera identifier identifying a second camera of the plurality of cameras as a second source camera different from the first source camera;

producing reference audio-video content according to the structured list of edit instructions;

generating one or more mates of the reference audio-video content by, for each of a plurality of selected camera cuts among the director-commanded camera cuts, modifying, in the structured list of edit instructions, the out-point time code of the corresponding first edit entry and the in-point time code of the corresponding following edit entry while retaining the first and second source-camera identifiers and the order of the first and second source cameras, such that, for each selected camera cut:

a transition from the first source camera to the second source camera occurs at a reference camera-switch timing in the reference audio-video content and at a later mate camera-switch timing in at least one of the one or more mates; and

in the at least one mate, selection of the first source camera is extended from the reference camera-switch timing to the later mate camera-switch timing and selection of the second source camera begins at the later mate camera-switch timing;

segmenting an ensemble comprising the reference audio-video content and the one or more mates into chunks;

generating a plurality of manifest files pointing to respective combinations of chunks selected from the ensemble, each respective combination of chunks preserving, for each of the plurality of selected camera cuts, a choice between the reference camera-switch timing and the later mate camera-switch timing of that selected camera cut, so that the respective combinations of chunks represent respective combinations of timing choices across the plurality of selected camera cuts;

delivering streamed audio-video content to respective recipients according to respective manifest files of the plurality of manifest files;

storing, in a ledger, associations between the respective manifest files and the recipients to which the streamed audio-video content was delivered according to the respective manifest files;

receiving the suspected unauthorized distribution;

applying a scene-change detection algorithm to the suspected unauthorized distribution to identify a plurality of time codes of camera cuts in the suspected unauthorized distribution;

building one or more reconstructed manifest files from the plurality of identified time codes, the one or more reconstructed manifest files including a reconstructed manifest file representing a detected combination of camera-switch timings across the plurality of selected camera cuts; and

searching the ledger to identify a recipient associated with a delivered manifest file that is equal to the reconstructed manifest file, the delivered manifest file that is equal to the reconstructed manifest file representing a recipient-associated combination of timing choices that includes at least one of the later mate camera-switch timings and matches the detected combination represented by the reconstructed manifest file.

36. (new) The method of claim 35, wherein identifying the plurality of time codes comprises comparing frames of the suspected unauthorized distribution with frames of one or both of the reference audio-video content and the one or more mates using perceptual hashes of the frames.

37. (new) The method of claim 36, wherein the comparing comprises fuzzy matching in which perceptual hashes of groups of frames are compared using sliding windows.

38. (new) A method of identifying a recipient associated with a suspected unauthorized distribution of audio-video content, the method comprising:

receiving the suspected unauthorized distribution;

applying a scene-change detection algorithm to the suspected unauthorized distribution to identify a plurality of time codes of camera cuts in the suspected unauthorized distribution;

building one or more reconstructed manifest files from the plurality of identified time codes, the one or more reconstructed manifest files including a reconstructed manifest file representing a detected combination of camera-cut timings across a plurality of camera cuts; and

searching a ledger comprising associations between a plurality of delivered manifest files and respective recipients to identify a recipient associated with a delivered manifest file that is equal to the reconstructed manifest file,

wherein each of the plurality of delivered manifest files identifies a respective combination of chunks selected from an ensemble comprising reference audio-video content and one or more mates of the reference audio-video content, and wherein, for each of the plurality of camera cuts, the reference audio-video content contains an ordered transition from a respective first source camera to a respective second source camera at a respective reference camera-cut timing, and at least one of the one or more mates contains the same ordered transition at a respective different mate camera-cut timing,

wherein each of the plurality of delivered manifest files represents, across the plurality of camera cuts, a recipient-associated combination of choices between the respective reference camera-cut timings and the respective different mate camera-cut timings, and

wherein the reconstructed manifest file represents, for at least one of the plurality of camera cuts, the respective different mate camera-cut timing detected in the suspected unauthorized distribution, and wherein the delivered manifest file that is equal to the reconstructed manifest file represents the same detected combination.